

## Final Project:

### Overview

The final lab project requires students to select one dataset from real-world to conduct comprehensive data analysis using data mining techniques. Each team, consisting of multiple members, will be responsible for customizing a specific number of data mining-related questions based on the dataset chosen. The number of **unique** questions to be customized will be equal to **the number of group members x 3**. These questions will encompass a variety of statistical and data mining methods, with up to three questions allowed to be of statistical nature, while the remaining questions will require the application of analytical models.

### Key Components of the Final Lab Project

1. **Selection of Dataset:** Students will choose one dataset from a provided list, ensuring it aligns with their interests and objectives for analysis.

2. **Customized Questions:** Each team will formulate a set of data mining-related questions based on the dataset chosen. These questions should cover various aspects of data analysis, including descriptive statistics, predictive modeling, and pattern recognition.

A. Statistical Questions: Up to **three questions** (for each group) can focus on statistical analysis, such as max/min/mean value testing or correlation analysis.

B. Data Mining Techniques: The remaining questions will require the application of data mining techniques, such as classification, clustering, or Association rules.

### Important:

*All data mining models must undergo proper model evaluation.*

This includes reporting metrics such as **Accuracy, Precision, Recall, F1-score, or Silhouette Score** (depending on the model type).

Each model should include both **quantitative evaluation** and a **short-written interpretation** of its performance.

3. **Visualization:** All problems in the final lab project must include visualization of the results with detailed decoration such as font size, color, text, and x-y labels. Visual representations, such as charts, or graphs, should be utilized to enhance understanding and interpretation of the data analysis outcomes.

4. **Final Report:** Teams will prepare a final report documenting their analysis process for each question. The report should include problem definitions, descriptions of analysis methods employed, visualizations of results, and highlights/takeaways from the analysis.

5. **Speech Presentation:** Each team member will present their assigned problem and research results during a speech presentation. Additional points will be awarded for effective communication and presentation skills.

**6. Submission of Report and Code:** Teams will upload the final report along with the original code used for analysis.

- **Gain Report:** Compile your findings, analysis, and visualizations into a comprehensive report. The report should be submitted as a PDF.
- **Code:** Include the entire code used for the analysis and upload them to GitHub.
- Submit your GitHub link to Canvas.

## Notes:

**Data mining** involves uncovering *hidden patterns, trends, and relationships* within datasets that are not immediately obvious. Your project should demonstrate an ability to explore data beyond simple descriptive statistics.

To receive full credit, your questions must reflect **depth and analytical insight**. Overly simplistic or surface-level questions (e.g., “What is the average value of X?”) will result in a **50% deduction**.

A simple example using the **Iris dataset** is provided for reference, but you are expected to design **original, creative, and meaningful data mining questions** relevant to your chosen dataset.

## Ideas for Meaningful Data Mining Questions

Here are some question types that go beyond surface-level analysis:

### 1. Correlation and Pattern Discovery

- *Example:* “Which combinations of student habits (study time, internet usage, absences) most strongly correlate with final grades?”
- *Idea:* Look for **hidden interactions** between multiple variables instead of single-variable summaries.

### 2. Classification and Prediction

- *Example:* “Can we predict whether a student will pass or fail based on social activities and family support?”
- *Idea:* Use **Decision Tree** or **Naive Bayes** models to classify outcomes and analyze feature importance.

### 3. Clustering and Segmentation

- *Example:* “Can we segment customers/students/users into meaningful clusters based on behavior or performance metrics?”
- *Idea:* Apply **K-Means** to find groups and then interpret what characterizes each cluster.

#### 4. Association Rules

- *Example:* “Which products or behaviors commonly occur together?”  
(e.g., “If a student spends more than 2 hours online and goes out frequently, are they likely to have lower grades?”)
- *Idea:* Use **Apriori** or **FP-Growth** to find frequent patterns or co-occurrences.

#### 5. Anomaly or Outlier Detection

- *Example:* “Which data points represent unusual or extreme behaviors, and what might explain them?”
- *Idea:* Use clustering distances or z-scores to detect anomalies.

#### 6. Temporal or Sequential Patterns

- *Example:* “How do student performance or user activities change over time?”
- *Idea:* Compare patterns across semesters, months, or sessions to find **temporal trends**.